NRSG 224 Patho Exam II (Classes 6-11): AI podcast scripts

Read-along text for the AI-voice audio. Content is drawn only from the professor's recorded lectures and class materials (slides, study guides, vocab lists).

Class 6. Circulatory Function, Part 1

The system and its job

The circulatory system has three components: the heart, which is the pump; the blood vessels, which carry blood to the tissues; and the blood, which fills the vessels. Its primary function is transport. It delivers oxygen and nutrients to the cells for energy. It carries waste products to the excretory organs, like the lungs and kidneys. It circulates electrolytes and hormones. It transports immune substances for defense. And it moves core heat to the skin, which helps regulate temperature. If even one of these fails, cells die. That's why a failing circulatory system causes so many problems.

Atria and ventricles

The heart has four chambers. Two atria on top, two ventricles on the bottom.

The atria have two jobs: they collect blood returning to the heart, and they act as pumps that help fill the ventricles. If the atria fail, two problems. First, the ventricles don't fill well, so less blood goes out per beat, and cardiac output drops. Second, blood pools and stagnates in the atria, and stagnant blood clots. Those clots can drain into the ventricles, get pumped out, and lodge in arteries elsewhere, causing ischemia and tissue damage.

The ventricles are the main pumping chambers. If they fail, cardiac output drops. Predisposing factors for impaired atrial or ventricular function: myocardial infarction and dysrhythmias, and also hypoxia or ischemia.

The right ventricle pumps blood to the lungs. That's the pulmonary circulation. If the right ventricle fails, blood doesn't get oxygenated, so the nursing problem is poor oxygenation.

The left ventricle pumps to the rest of the body. That's the systemic circulation. If the left ventricle fails, oxygenated blood doesn't reach the tissues, so the problem is decreased systemic perfusion, and major organs can fail.

Cardiac cycle and cardiac output

Systole: the heart contracts and pumps blood out. Diastole: the heart relaxes and fills.

Cardiac output, C O, is the amount of blood the heart pumps per minute. It measures how well the heart works as a pump. Formula: cardiac output equals heart rate times stroke volume. Beats per minute, times blood per beat, gives blood per minute.

Heart rate: up to a point, a faster rate raises cardiac output. But at very high rates, roughly 150 to 200 depending on fitness, the heart spends less time in diastole, so the ventricles don't have time to fill, stroke volume falls, and cardiac output drops. A rate that's too slow also drops cardiac output. Heart rate is set mainly by the autonomic nervous system, the balance of sympathetic versus parasympathetic activity.

Stroke volume: preload, afterload, contractility

Stroke volume is the blood pumped out with each beat. It depends on the loading conditions, preload and afterload, plus myocardial contractility.

Preload is the amount of blood in the ventricle ready to be pumped out on the next beat. It's set by venous return. More venous return means more blood in the ventricle, more stretch of the muscle fibers, a stronger contraction, and a bigger stroke volume. That's Starling's law of the heart, also called the Frank-Starling mechanism. Think of an elastic band: more stretch, harder snap back. But there's a limit. Maximum force, and maximum cardiac output, happens when the fibers are stretched about two and a half times their resting length. Underfilling, or overfilling beyond that, drops cardiac output. Picture an overstretched waistband that won't snap back.

Afterload is the resistance the heart must pump against to eject blood. It's determined mainly by arterial blood pressure. Higher pressure, the harder the heart works, the lower the stroke volume and cardiac output. Think of pumping a bike tire: a flat tire takes lots of air per stroke, a full one takes less. Afterload also rises with narrowing, or stenosis, of the aortic or pulmonic valve.

Cardiac contractility is the force of ventricular contraction independent of preload and afterload. It comes from actin and myosin interaction. Factors affecting it are called inotropic influences. A positive inotrope increases force. Example: sympathetic stimulation, where the heart pounds and stroke volume rises. A negative inotrope decreases force. Examples: hypoxia or ischemia, because the muscle can't make enough A T P to contract; and a heart attack, because the dead segment of wall doesn't contract with the rest.

Why it matters: without cardiac output, none of the transport functions happen. Most cardiac diseases cause problems by dropping cardiac output, and interventions aim to raise it: increase preload, decrease afterload, or increase contractility.

Blood vessels and blood pressure

Three kinds of vessels: arteries carry blood from the heart to the tissues, capillaries are where gases and nutrients exchange, and veins return blood to the heart.

Every vessel except capillaries has three layers. The outer tunica externa is fibrous connective tissue for support. The middle tunica media is smooth muscle that constricts and relaxes to control diameter and match flow to tissue needs. The inner tunica intima is endothelium: a smooth, slippery surface that keeps platelets from sticking, and it also makes vasoactive substances that affect vasoconstriction, vasodilation, and platelet aggregation.

Blood flow needs two things: patent vessels, and a force to move the blood. That force is arterial blood pressure. Normal is about 120 over 80. If pressure is too low, there's no driving force and circulation is poor.

Key formula: blood pressure equals cardiac output times peripheral vascular resistance, P V R. More cardiac output means higher pressure, like stuffing more people into a phone booth. Factors affecting cardiac output: heart rate, contractility, venous return, and blood volume. Blood volume works through venous return, and it's the easiest one to change, with I V fluids or diuretics.

Peripheral vascular resistance is the resistance blood meets as it moves through the vessels. The big factor is vessel diameter, set by arterial constriction and elasticity. Narrower vessel, more resistance, higher pressure. Dilated vessel, lower pressure.

The factors easiest to change with drugs are blood volume, heart rate, and vessel diameter. Blood pressure too low: give I V fluid to raise volume, or a vasoconstrictor to raise resistance. Blood pressure too

high: a diuretic to lower volume, a drug to slow the heart rate, or a vasodilator to lower resistance. That's why the blood pressure drug classes make sense.

Control mechanisms: A N S, kidneys, A D H

Three main controllers.

One: the autonomic nervous system. It works through two centers in the medulla. The vasomotor center produces sympathetic effects: increased heart rate and contractility, and changes in vessel tone. The cardioinhibitory center produces parasympathetic effects: decreased heart rate. These centers get input from other brain areas like the hypothalamus, and from arterial baroreceptors and chemoreceptors. The A N S affects heart rate and vessel diameter, but not blood volume.

Two: the kidneys. They adjust sodium and water excretion to keep blood pressure normal, and they run the renin-angiotensin-aldosterone system, R A A S. When blood pressure falls, the juxtaglomerular cells of the kidney release renin. Renin converts angiotensinogen to angiotensin one. A C E, in the blood vessels of the lung, converts angiotensin one to angiotensin two. Angiotensin two is a powerful vasoconstrictor, the short-term mechanism, and it signals the adrenal cortex to release aldosterone, which retains sodium and water, the long-term mechanism. Result: blood pressure rises. Kidneys affect volume and vessel diameter, not heart rate.

Three: A D H, antidiuretic hormone, also called vasopressin. It's released from the posterior pituitary when blood volume or pressure is low. Its antidiuretic effect retains water and raises volume; its vasopressor effect constricts vessels. Volume and diameter, not heart rate.

Hook: A N S does rate plus diameter. Kidneys and A D H do volume plus diameter.

Hypertension

Primary, or essential, hypertension has no identifiable disease causing it. Secondary hypertension has an identifiable cause, like kidney or thyroid disease, and if you remove the cause, the pressure should come down.

In primary hypertension, cardiac output and or peripheral resistance are increased, related to altered sympathetic activity and or altered kidney regulation. The three control systems seem to be reprogrammed to hold pressure at a higher set point. It's not a disease process of one system.

The 2017 guidelines. Normal: systolic under 120 and diastolic under 80; both must be normal. Elevated: 120 to 129 and diastolic under 80. Stage one: 130 to 139, or 80 to 89. Stage two: 140 or higher, or 90 or higher. Stages use "or," so only one number has to fit. If the two numbers fall in different categories, use the higher one. 140 over 90 used to be the cutoff for diagnosis; now it's already stage two. Pre-hypertension was dropped because people heard "pre" and thought they didn't have it yet.

Diagnosis requires elevated readings on two or more separate occasions, averaging two or more careful readings each time. One high reading isn't enough.

Hypertension is the silent killer: usually no symptoms except the number. Very high pressure may cause headaches or nosebleeds.

Why we care: it predisposes to atherosclerosis, heart disease, and stroke, two ways. One, mechanical stress makes the arterial wall hypertrophy, so it gets thicker, narrower, and less elastic, hardened. Two, the stress injures the endothelium.

Predisposing factors. Non-modifiable: family history; older age, though it's showing up younger as lifestyle risks rise; race, highest in Black or African American patients, more frequent, more severe, and earlier, cause unclear, possibly salt sensitivity or higher sympathetic activity; and sex, men higher until menopause, then women are equal.

Modifiable: high sodium intake, which holds water. Low potassium intake; the sodium-to-potassium ratio may matter. Don't give potassium supplements, since overdose causes hyperkalemia and dysrhythmias; encourage potassium-rich fruits and vegetables instead. Obesity, especially upper-body or central obesity, the apple shape. Belly fat acts like an endocrine organ, affecting pressure, glucose, and inflammation. Measure waist circumference: risk if over 35 inches in women, over 40 in men. Obesity also contributes to sleep apnea, and poor sleep raises risk. Physical inactivity, even apart from obesity. Excess alcohol: more than two drinks a day for men, more than one for women, and you can't save them up for the weekend; that's binge drinking. Possibly stress, which is chronic and social, like poverty, abuse, racism. Smoking and high cholesterol relate more to heart disease than to hypertension.

Nursing problem: persistently increased blood pressure with long-term vascular damage and risk of atherosclerosis.

Interventions: everyone with elevated, stage one, or stage two pressure gets lifestyle modification: weight loss, less sodium and more potassium, more activity, less alcohol. Stage one with few risk factors may stay on lifestyle alone. Stage two, and some stage one, get antihypertensives plus lifestyle. If one drug fails, add one with a different target: volume, heart rate, or diameter. Goal: taper meds as lifestyle works. Patient education is key, because there are no symptoms but the drugs have side effects, so explain why, and help fit changes into their life.

Endothelium and atherosclerosis

Arteriosclerosis is the general term for hardening and thickening of arteries, including calcification. Atherosclerosis is the specific form: fibro-fatty plaque with a lipid core and fibrous cap in the intimal lining. Don't mix them up.

The process. Step one: injury to the endothelium, from hypertension's mechanical stress, smoking's free radicals, or inflammation from pathogens. Step two: macrophages stick to the injured area and ingest L D L cholesterol, becoming foam cells. Foam cells squeeze under the lining and die, releasing fat and cholesterol: fatty streaks. Step three: the injured endothelium makes less antithrombotic and vasodilating substance and releases growth factors, so smooth muscle proliferates and forms a fibrous cap. That's the plaque.

Three problems from plaque. One, it narrows the lumen and reduces flow. Two, endothelial damage promotes clotting. Three, the cap can rupture when fat builds pressure inside; platelets gather, a clot totally occludes the artery, and you get an M I or ischemic stroke. Pieces can also embolize downstream.

It's slow. Fatty streaks appear in young adults, but symptoms may not show for 20 to 40 years.

Predisposing factors. Non-modifiable: age, male sex or postmenopausal female, and family history of premature coronary heart disease, meaning before 45 in men or before 55 in women. Modifiable:

smoking, hypertension, and diabetes, which all injure the endothelium, and hypercholesterolemia, the primary risk factor. Also possibly obesity, inactivity, stress, and poor sleep.

Cholesterol

Cholesterol is a fat-like substance found only in animals, not plants. It's essential for cell membranes, bile salts, and steroid hormones; it's bad only in excess. It doesn't dissolve in blood, so it travels in lipoproteins. L D L is the main carrier of cholesterol to the tissues; whatever isn't used is available for plaque. Want it low. H D L clears cholesterol from blood and plaques and returns it to the liver to be excreted in bile. Want it high. Hook: H D L is Healthy and High; L D L is Lousy and Low.

Goals: total cholesterol under 200, with some experts aiming for 170 or even 150 in high-risk patients. L D L under 100, some aim for 70. H D L over 40; 60 or more is protective. Triglycerides under 200, some aim for 150.

Ischemia, infarction, collaterals

Ischemia: arterial flow isn't enough to meet oxygen demand, but no permanent damage yet. Infarction: an area of ischemic necrosis, dead tissue. It's a matter of degree. Findings depend on which artery and how blocked.

Collateral circulation: alternate small vessels that widen to supply an area if the main artery is blocked. Found in the heart, brain, and large leg muscles. Collaterals develop with gradual blockage, which is why atherosclerosis is often silent until plaque is large. A sudden clot leaves no time, so you get an acute problem.

Interventions: lifestyle first: stop smoking, control diabetes, hypertension, and cholesterol, with meds if needed, then taper.

Coronary heart disease

Coronary heart disease, also called coronary artery disease or atherosclerotic heart disease, C H D, C A D, or A S H D. She prefers C A D because it names the location. Heart muscle gets its oxygen only from the coronary arteries, not from blood inside the chambers. Causes of obstruction: atherosclerosis, the most common; vasospasm; and disruption of a plaque causing thrombosis. Predisposing factors are the same as for atherosclerosis.

The nursing problem is oxygen supply less than demand: ischemia and chest pain, and with an M I, cell death and decreased cardiac output.

Three presentations: asymptomatic, chronic ischemic heart disease, which is stable and variant angina, and acute coronary syndrome, which is unstable angina and acute M I.

Angina pectoris is chest pain or pressure from ischemia, without cell death. The clogged arteries can't meet demand, cells switch to anaerobic metabolism, and lactic acid is thought to cause the pain. It usually doesn't start until the artery is about 75 percent occluded, because collaterals compensate. Patients say pressure or tightness, "an elephant on my chest," and may deny pain, so ask about discomfort. Location: substernal or precordial, may radiate to the left shoulder and down the inside of the left arm. Levine's sign is a clenched fist over the chest. Atypical radiation: to the right, the jaw, the epigastric area, or the back. Women are less likely to have classic symptoms; in the month before an M I they report unusual fatigue,

shortness of breath, nausea, and dizziness, which delays recognition and worsens prognosis.

Stable angina, also called classic or exertional: triggered by increased workload like exertion, cold, anger, or a big meal. Predictable: same intensity, same trigger, same duration. Relieved by rest or nitroglycerin. Cause: a stable, unruptured plaque.

Variant angina, also called vasospastic or Prinzmetal's: coronary vasospasm, not plaque. Usually at rest or during sleep. Cause of spasm unknown. Least common type.

Unstable angina, also called preinfarction angina: more frequent, more intense, longer, or not relieved as usual. Cause: a ruptured plaque with a clot that breaks down before heart cells die. Cardiac markers do not rise. It often precedes an M I, and in the E R it's treated like an M I until ruled out.

Acute myocardial infarction: usually a ruptured plaque and clot that does not break down, so cells die. Zones: dead, injured, and ischemic. Findings: pain more severe and longer, not relieved by rest or nitroglycerin, needing opioids. Nausea and vomiting. Sympathetic response: tachycardia, pale cool clammy skin, restlessness, anxiety, a feeling of impending doom. Dysrhythmias from damaged cells. The dead wall doesn't contract, so cardiac output and blood pressure drop, possibly heart failure, or shock if large. Inflammation causes fever and increased white count. Cardiac markers leak from dead cells: C K, L D, myoglobin, and troponin, the gold standard.

C A D interventions

The problem is supply versus demand.

Decrease demand: pace activities; drugs to lower heart rate or blood pressure; prevent vasoconstriction by quitting smoking, avoiding stress, and keeping warm.

Nitroglycerin, a white pill in a brown bottle, under the tongue. It's a vasodilator, but it works by decreasing demand, not by opening the hardened coronaries. It dilates the other arteries, lowering pressure and afterload, and it dilates veins, lowering venous return and preload, so the heart works less. Three-by-five rule: up to three doses, five minutes apart. Still pain after three, treat it as an M I and go to the E R.

Increase supply: only by revascularization. Angioplasty, P T C A: a balloon catheter, usually through the femoral artery, squashes the plaque, and a stent keeps the artery open. Risks: artery damage and knocking plaque loose. Coronary artery bypass graft, CABG: a leg vein or synthetic graft bypasses the blockage. More invasive, but better for multiple blockages, like a triple or quadruple bypass.

Low-dose aspirin: an antiplatelet, reducing platelet aggregation to prevent a first or second M I. Avoid with ulcers, aneurysms, or bleeding disorders. It's no longer automatic; weigh benefit against bleeding risk.

For M I only: thrombolytic, or fibrinolytic, therapy breaks down the clot. That's different from anticoagulants like heparin, which only prevent clots from growing or forming. The goal is to reopen the artery and save injured cells; dead cells can't be saved. Give within 12 hours of symptom onset, best within 2 hours. Not for patients at high risk of bleeding. Also for M I only: opioids like morphine, because severe pain drives sympathetic stimulation and tachycardia, which raises workload.

Test tip: when asked why a drug works, map it to heart rate, preload, afterload, contractility, volume, or diameter.

Rapid recap

Cardiac output equals heart rate times stroke volume. Stroke volume depends on preload, afterload, and contractility. Blood pressure equals cardiac output times resistance. A N S controls rate and diameter; kidneys and A D H control volume and diameter. Normal is under 120 and under 80; stage two is 140 or 90. Atherosclerosis starts with endothelial injury, foam cells, fatty streaks, and a fibrous cap. H D L high, L D L low. Stable angina is predictable and relieved by rest or nitro. Variant is spasm at rest. Unstable is a ruptured plaque, worse pain, no marker rise. M I means dead cells, markers up, troponin is gold standard. Nitro lowers demand; angioplasty and bypass increase supply; thrombolytics within 12 hours, only for M I.

Class 7. Circulatory Function, Part 2

Heart valves

Valves keep blood flowing one way. The atrioventricular, or A V, valves sit between atria and ventricles: the tricuspid on the right, the mitral, or bicuspid, on the left. The semilunar valves control flow out of the ventricles: the pulmonic, into the pulmonary artery, and the aortic, into the aorta. There are no valves at the entrance to the atria, so when the atria overfill, as in heart failure, blood backs out. That's why you see jugular vein distention in right heart failure and lung congestion in left heart failure.

Pathological change: valves become stiff or deformed. Predisposing factors: congenital defects, and scar tissue from trauma, ischemia, or inflammation.

Valvular stenosis means narrowing. The valve doesn't open fully, so the chamber behind it has to push harder through a tiny opening. Over time that chamber hypertrophies and can fail.

Valvular incompetence, also called regurgitation, means the valve doesn't close fully, so blood leaks backward and has to be pumped again. The receiving chamber gets volume overloaded.

Bottom line for both: the heart's workload goes up and cardiac output goes down. Mild valve disease may show no signs of low output, but you'll hear a heart murmur, the sound of turbulent flow, blood forced through a narrow opening or flowing backward.

Pericardium and cardiac tamponade

The pericardium has two layers. The tough outer fibrous layer doesn't stretch. The inner serous layer lines the heart and the outer layer, forming a cavity with only about 30 to 50 milliliters of fluid, two or three tablespoons. Functions: it holds the heart's position in the chest, lubricates so the layers don't rub, acts as a barrier to infection, and prevents overfilling of the heart.

Pathological change: excess fluid in the cavity, a pericardial effusion. The fluid can be serous, pus, or blood. Predisposing factors: trauma, inflammation, infection, cancer, and myocardial infarction with rupture of the heart wall.

Cardiac tamponade: the fluid compresses the heart so the chambers can't fill. Preload falls and cardiac output drops. It's an emergency. A rapid effusion causes tamponade faster than a slow one, because the tough outer layer can't stretch quickly.

Myocardium and cardiomyopathy

The myocardium does the work of the heart. Its contractions are involuntary. Pathological change: disease of the heart muscle, cardiomyopathy. Predisposing factors: genetic defects, other heart disease, viral infection, ischemia, autoimmune and collagen diseases, drugs or chemicals, radiation, and metabolic diseases. Myocardial infarction also damages it, because dead muscle doesn't contract. Damaged myocardium can't generate normal force, so stroke volume and cardiac output fall.

Endocardium and infective endocarditis

The endocardium lines the heart. It's smooth and slippery so clots, platelets, and microorganisms don't stick.

Infective endocarditis: pathogens invade the endocardial surface and form vegetative lesions. Lecture definition: microorganisms and cellular debris tangled in strands of fibrin. Not little plants.

Predisposing factors come in two parts. First, an environment that lets organisms grab on: preexisting heart defects, valve disease, prosthetic valves, damage to the endocardium, and implanted pacemakers, defibrillators, or central lines. Second, a portal of entry into the blood: an oral lesion, dental procedure, upper respiratory infection, skin wound, food poisoning, or a gum injury. Organisms reach the heart through the blood.

Three worries with vegetative lesions. One, blood picks up pathogens passing the lesion and spreads infection elsewhere. Two, the lesion destroys the underlying valve tissue, so valve damage gets worse. Three, a piece can break off as an embolus, block a smaller artery, and cause infarction or stroke.

Prophylaxis means prevention. High-risk patients, like those with prosthetic valves, may get antibiotics before procedures that introduce bacteria, like dental work. A heart murmur on the history means evaluate for prophylaxis; it's not given in every case. Tell any provider about a heart defect before an invasive procedure.

Heart failure basics

Heart failure is failure of the heart as a pump: cardiac output is too low to meet the body's needs. It's not cardiac arrest; the heart is still beating. Congestive heart failure, C H F, means fluid accumulates because blood backs up when it isn't pumped forward.

Predisposing factors: impaired cardiac function, like M I, cardiomyopathy, or valve disease; and excessive work demands, like hypertension or fluid overload.

Compensatory mechanisms help short term but hurt long term because they raise workload.

One, increased sympathetic activity raises heart rate and contractility, which raises workload.

Two, R A A S activation retains sodium and water, raising volume, and vasoconstricts, raising afterload.

Three, myocardial remodeling, hypertrophy. A thicker left ventricular wall shrinks the chamber so it fills less, and it needs more oxygen, which can cause ischemia if the coronaries are narrowed.

The one helpful long-term mechanism: natriuretic peptides. A N P, atrial natriuretic peptide, comes from the atria. B N P, B-type or brain natriuretic peptide, comes from the ventricles; it's called brain because it was first found in pig brain. They're released when the chambers are stretched by volume overload. Natriuretic means excreting sodium in urine. Effects: decreased sympathetic activity, decreased sodium and water retention, and decreased vasoconstriction. Net: lower volume and lower afterload, opposing the S N S and R A A S. B N P blood levels are checked in the E R: higher B N P means more severe heart failure. Synthetic B N P is I V only, usually in I C U.

Ejection fraction: H F r E F versus H F p E F

Ejection fraction is the percent of blood ejected from the ventricle when it contracts. Normal is about 50 to 75 percent, measured by ultrasound.

Systolic heart failure, heart failure with reduced ejection fraction, H F r E F: the ventricle doesn't empty well because contraction is weak. Examples: cardiomyopathy and M I.

Diastolic heart failure, heart failure with preserved ejection fraction, H F p E F: the ventricle doesn't fill well. Examples: chronic hypertension, where the thick wall shrinks the chamber, and cardiac tamponade, where the ventricles are compressed. Some patients have both.

Right versus left heart failure

Draw the path: venous system, right heart, lungs, left heart, arterial system. Failure causes backward effects behind the failing side and forward effects ahead of it. She won't ask you to label symptoms forward or backward; it's just a way to group them.

Right heart failure. Forward: blood doesn't reach the lungs to get oxygen, so hypoxia: shortness of breath, tachycardia, anxiety; and less blood reaches the left heart, so systemic output falls. Backward: blood backs into the veins. Jugular venous distention, since neck veins are normally flat. Then fluid is pushed into tissues: dependent edema, which depends on position, the sacrum if lying flat. Peripheral edema in legs, ankles, and feet. Hepatomegaly, with right upper quadrant pain. Splenomegaly. G I congestion, with poor absorption and appetite. Ascites, fluid in the abdomen. Weight gain: one pound is about 500 milliliters, so two pounds overnight is about a liter of fluid, not fat.

Left heart failure. Forward: poor tissue oxygenation, so fatigue, weakness, and poor activity tolerance; a sympathetic response with tachycardia and pale, cool, clammy skin; and if output drops severely, low blood pressure, poor organ perfusion, and cardiogenic shock, shock originating from the heart, which can lead to multi-organ failure and death. Backward: blood backs up into the lungs, fluid is pushed into the alveoli, and that's pulmonary edema. Gas exchange fails, causing hypoxia, moist cough, and crackles. Severe pulmonary edema is like drowning in your own fluid.

Crackles are the sound of air moving through fluid. Fine crackles are fluid in smaller airways, like Rice Krispies in milk. Coarse crackles are fluid in larger airways, like slurping the bottom of a straw.

Orthopnea is shortness of breath lying flat, relieved by sitting up. Upright, gravity pools blood in the legs. Lying flat sends it back to the heart, where it gets stuck in the lungs, worsening pulmonary edema. Paroxysmal nocturnal dyspnea is the same mechanism at night.

Compare: both sides cause hypoxia. Only left causes pulmonary edema; right causes peripheral edema. And once one side fails, the other follows: if the right fails, the left has nothing to pump; if the left fails, pressure backs up in the lungs and overworks the right. Catch it early.

Highest-priority problem: poor oxygenation if hypoxia is severe, or decreased cardiac output if it's bad enough to cause shock.

Heart failure treatment

First, find and fix the cause if possible, like controlling hypertension or fluid overload. Some causes, like dead muscle after an M I, can't be fixed.

Four drug goals. One, increase contractility, with a positive inotrope like digoxin. Two, decrease edema, with diuretics and sodium restriction. Three, decrease the heart's workload, with drugs that lower volume, heart rate, or blood pressure. Four, counteract the S N S and R A A S. Monitor carefully so output doesn't

get worse. End stage: heart transplant, but donors are limited.

Arteries and capillaries

Arteries and arterioles carry blood to tissues. They have thick, elastic walls because pressure is high. Three layers, except capillaries: tunica externa for support, tunica media of smooth muscle controlling diameter, and tunica intima, the endothelium, which prevents platelet sticking and makes vasoactive substances.

Pathological change: impaired function. Predisposing factors: atherosclerosis, obstruction by an embolus or thrombus, and trauma. Nursing problems: ischemia, flow too low for oxygen demand, or infarction, ischemic necrosis.

Capillaries are the exchange site for gases, nutrients, and wastes. Their wall is one endothelial layer, and they're so narrow that red cells pass single file. About 10 billion capillaries give a surface area of about 100 football fields. Predisposing factors for capillary problems: trauma and decreased blood supply. Symptoms depend on which tissue they supply.

Chronic arterial impairment: P A D

Peripheral artery disease, also called atherosclerotic occlusive disease or arteriosclerosis obliterans. Gradual narrowing, usually atherosclerosis, so collateral circulation develops and symptoms appear late.

Findings: intermittent claudication, ischemic muscle pain with walking relieved by rest, often charted as "one-block claudication." Paresthesias, abnormal sensations from ischemic nerves. Cool, pale extremity; weak or absent pulses; slow capillary refill; hair loss; thin, shiny skin; thick, brittle toenails. Pain when the legs are elevated, because blood must flow uphill. Arterial ulcers are punched out, sharply edged, pale, and dry. If infarction occurs, there's eschar, dead tissue that must be debrided to heal. A dry necrotic toe that could break off developed slowly; a necrotic area that's still wet developed quickly.

Acute arterial occlusion: the seven P's

Usually a clot, embolus or thrombus, so it's sudden with no time for collaterals. An emergency. The seven P's: pallor, polar meaning cold, pulselessness, pain, paresthesias, paralysis, and pistol shot, meaning sudden onset like a bullet. Same symptoms as chronic, but fast and severe: cold, not cool; pulses absent, not weak.

Chronic versus acute, arterial or venous: chronic develops slowly, starts mild, worsens over time, and you keep it for life. Acute comes on suddenly and severely, and is either fatal or goes away.

Veins and chronic venous insufficiency

Veins and venules return blood to the heart and serve as storage. Thin, stretchy, low-pressure walls, holding about two-thirds of the blood volume. Two things move blood uphill: venous valves and the skeletal muscle pump, where contracting muscles squeeze the veins and valves prevent backflow. Immobility removes the muscle pump, so blood pools. Pooled blood is venous stasis.

Predisposing factors for venous impairment: incompetent valves, immobility, obstruction by embolus or thrombus, and trauma. Incompetent valves happen when a stretched vein pulls the leaflets apart. Immobility is the most common cause, and preventing its complications is the nurse's job.

Chronic venous insufficiency, from long-standing stasis and valve failure: the foot is warm and pink to red, pulses strong, capillary refill fast, with edema, usually both legs. Aching worsens with standing, and elevating the legs helps, the opposite of arterial. Venous stasis ulcers, usually at the medial ankle, are red, wet, and beefy, like raw hamburger. Severe venous problems can slow arterial flow too, and one leg can have both.

Acute venous impairment: D V T

Deep vein thrombosis, also called thrombophlebitis: a clot in a deep vein that irritates and inflames it. Signs: redness, heat, pain, and swelling, usually in one leg. The D V T leg is hot; chronic venous insufficiency is only warm.

Virchow's triad, three factors that promote clots: venous stasis, hypercoagulability, and injury to the vessel wall. Hypercoagulability example: dehydration or a blood disorder. Wall injury: trauma or an I V in that vein. Stasis from immobility is most common, and nurses prevent it. The big risk: the clot embolizes to the lungs, a pulmonary embolism.

Arterial versus venous logic

Don't memorize swappable lists. Arterial means blood can't get in: pale, cool or cold, weak or absent pulses, slow refill, dry punched-out ulcers, worse when elevated. Venous means blood can't get out: warm, pink to red, strong pulses, fast refill, edema, wet red ulcers, better when elevated. And compare right versus left failure, acute versus chronic. Understand why each causes its symptoms.

Rapid recap

Stenosis doesn't open; regurgitation doesn't close; both raise workload and drop output, with a murmur. Tamponade compresses the heart so it can't fill. Endocarditis needs a place to grab on plus a portal into the blood; vegetative lesions spread infection, destroy valves, and embolize. Heart failure compensations, S N S, R A A S, remodeling, help short term and hurt long term; A N P and B N P help. Reduced ejection fraction is systolic; preserved is diastolic. Right failure backs up into the body: J V D, edema, liver. Left failure backs up into the lungs: pulmonary edema, crackles, orthopnea, and forward into cardiogenic shock. P A D gives claudication; acute occlusion gives the seven P's. Venous is warm, swollen, better elevated. D V T is a hot, swollen leg; Virchow's triad is stasis, hypercoagulability, and wall injury.

Class 8. Circulatory Part 3 and Respiratory Part 1

Test scope for dysrhythmias

She will not show rhythm strips or ask you to identify specific dysrhythmias; that's critical care. She will ask about basic E K G elements, predisposing factors for dysrhythmias, why we care, and interventions.

Conduction system

Heart muscle cells can generate their own action potentials. Certain cells do it faster, and they form the conduction system. The S A node, sinoatrial node, is the primary pacemaker at 60 to 100 beats per minute and controls the atria. The A V node, atrioventricular node, connects atria and ventricles and delays the impulse so the atria can empty. Its backup rate is 40 to 60. Then the bundle of His, also called the A V bundle, the right and left bundle branches, and the Purkinje fibers, whose backup rate is 15 to 40. Hook: 60, 40, 15. Fast conduction through this system makes the ventricles contract nearly together, which maximizes filling, stroke volume, and cardiac output. If the A V node is blocked, atria and ventricles beat independently.

Automaticity: heart cells generate their own action potentials without nervous-system input. Mechanism: S A and A V node cells have a constant slow leak of sodium inward, so they depolarize spontaneously.

Conductivity: the impulse's ability to travel cell to cell.

Action potential and refractory periods

At rest, the inside of the cell is more negative; potassium is high inside and sodium high outside. In an action potential, permeability changes, sodium rushes in, and the cell depolarizes.

Absolute refractory period: no new action potential can fire, no matter how strong the stimulus. It's longer in heart muscle than skeletal muscle so a second contraction can't start before the first ends. That prevents sustained contraction, which would be cardiac arrest; the heart has to relax to fill.

Relative refractory period: a new action potential fires only with a much larger than normal stimulus.

Supernormal excitatory period: even a weak stimulus can trigger one. Abnormalities in the relative or supernormal periods predispose to dysrhythmias.

E K G basics

An E K G, or E C G, electrocardiogram, records electrical activity. Horizontal axis is time; vertical is amplitude. P wave: depolarization of the S A node and atria. Q R S complex: depolarization of the ventricles. T wave: repolarization of the ventricles. Atrial repolarization is hidden inside the Q R S.

Normal sinus rhythm needs a P wave before every Q R S, and constant R-R intervals. The R-R interval is the distance between peaks of consecutive Q R S complexes. Uneven spacing means an irregular rhythm.

Dysrhythmias

Dysrhythmia means bad rhythm and is her preferred term. Arrhythmia, technically "without rhythm," means the same thing in textbooks. Why we care: they can drop cardiac output, by changing rate, filling time, or the atria-then-ventricles timing. Atrial kick is the strong atrial contraction that fully empties the atria into the ventricles just before the ventricles contract. Losing that timing loses the atrial kick and drops output.

Three mechanisms. One, automaticity problems: rate too fast or too slow, or premature ectopic beats. Two, conductivity problems: delayed or blocked conduction from scar tissue, like after an M I, or inflammation. In a bundle branch block, parts of the ventricle depolarize at different times, so contraction weakens. Three, reentry: normally the impulse dies out because all cells become refractory. With a delay or block, the impulse comes through late and re-enters tissue that has already repolarized, depolarizing it again. Same tissue, same impulse, twice. That ruins the timing and drops filling and output.

Premature ectopic beat: an impulse from an abnormal location, not the S A node. It has to be premature to steal the pacemaker role. Once it does, the S A node has trouble taking back over because surrounding cells are still refractory. A premature ventricular contraction, P V C, comes before the halfway point between two normal beats and has a wide, bizarre Q R S because it didn't use the fast pathway. One occasional P V C in a healthy heart probably doesn't matter; many can predispose to worse dysrhythmias.

Predisposing factors, which she will ask: a congenital conduction defect; later damage to the conduction system, like cardiomyopathy or M I scar; inflammation of the heart, which changes refractory periods and conductivity; electrolyte imbalances, which change sodium and potassium gradients; and medications, prescription or over the counter, herbals, and recreational drugs.

Examples

Sinus bradycardia: rate under 60. Sinus tachycardia: over 100. Bradycardia means fewer beats; tachycardia means less filling time. Both can be normal: tachycardia with exercise, bradycardia in very fit athletes with a strong stroke volume.

Atrial fibrillation, A fib: impulses bounce randomly through the atria, which quiver. The rhythm is very irregular, so count the pulse a full minute. On the E K G, no distinct P waves, a scribbly baseline. Two problems: poor ventricular filling and lost atrial kick, so output drops; and blood pools in the atria, forms clots, and causes emboli. A fib is one of the most common causes of ischemic stroke, so stroke workups include an E K G.

Heart block, or A V block. First degree: every P has a Q R S, but the P to Q R S interval is longer. Second degree: some P waves have no Q R S. Third degree, complete: P waves evenly spaced and Q R S complexes evenly spaced, but unrelated; atria and ventricles beat independently.

Ventricular tachycardia: 100 to 250 a minute, up to about four beats a second. No time to fill, no atrial kick, and output really drops.

Ventricular fibrillation: the ventricles quiver with no effective contraction. Output is essentially zero. The E K G is chaotic with no identifiable Q R S.

Fibrillation in general means quivering, disorganized depolarization.

Assessment and interventions

Too slow or too fast both drop output. Signs: mild, fatigue and poor activity tolerance; more severe, dizziness, fainting, low blood pressure; worst, heart failure or shock. Also weak pulses, cool clammy skin, altered mental status, chest pain, and dyspnea. Very fast rates raise oxygen demand and can trigger an M I if the coronaries are narrowed.

Intervention order: first find and remove the cause, like fixing electrolytes. If you can't, use anti-dysrhythmic drugs or electrical interventions. The drugs, potassium, sodium, and calcium channel blockers, change membrane permeability, which changes automaticity, conductivity, and refractory periods. That's all she wants you to know.

Electronic pacemaker: a generator under the chest skin with leads against the walls of the atria and ventricles. It senses the rate and fires if it drops below a set level. Used for certain A V blocks and bradydysrhythmias.

Defibrillation: a much bigger shock than a pacemaker's, to depolarize all heart cells at once, stop the chaos, and let the S A node regain control. Used for V fib and pulseless rhythms. An A E D shocks only shockable rhythms. Cardioversion is the other electrical option.

Respiratory overview and conducting airways

The primary function is gas exchange: oxygen in, carbon dioxide out. If it fails, you get hypoxia, not enough oxygen to the tissues, and hypercapnia, too much carbon dioxide.

Conducting airways warm, moisten, and filter air. No gas exchange happens there. Upper: nasal passages, mouth, pharynx, larynx. Lower: trachea, bronchi, terminal bronchioles. Three layers: cartilage rings in larger airways to keep them from collapsing; criss-crossing smooth muscle to control diameter; and mucosa with mucus cells and cilia. The mucociliary blanket, or escalator: mucus traps particles, and cilia move them up and out.

Problems. Bronchospasm, from allergic reaction or electrolyte imbalance, narrows the airway and raises resistance. An impaired mucociliary blanket, from smoking or dry air, leaves secretions and irritants behind, causing infection and inflammation. Mucosal swelling, from injury, infection, or allergy, narrows the lumen. Nursing problems: impaired airflow, and risk of infection.

Alveoli

Gases cross the alveolar-capillary membrane: oxygen from alveolus to blood, carbon dioxide the other way.

Type one alveolar cells are thin epithelial cells next to the capillary; they're the site of gas exchange. Damage means impaired gas exchange.

Type two cells make surfactant. Surface tension pulls alveoli shut; surfactant interrupts it so they stay open. Surfactant production is impaired by lack of deep breathing, toxic gases, anesthesia, and oxygen toxicity. Less surfactant means more surface tension and atelectasis.

Alveolar macrophages clear debris and pathogens. Smoking or immune disorders impair them, raising infection risk.

Atelectasis is collapse of alveoli, anywhere from a few to a whole lobe. Patients lying flat breathe shallowly, and you hear decreased breath sounds over the collapsed area. Deep breaths stimulate surfactant. Prevention: turn, cough, and deep breathe, plus the incentive spirometer, ten times an hour while awake.

Pleura

The parietal pleura lines the chest wall; the visceral pleura covers the lung. Between them is the pleural space, a potential space with a little serous fluid to reduce friction, kept at negative pressure, a vacuum that keeps the elastic lung from recoiling.

Pleuritis, or pleurisy: inflammation of the pleura, often from lung infection spreading. It causes pleuritic chest pain: sharp, stabbing, worse with a deep breath or cough, often one-sided. Cardiac pain is pressure; sore muscles from coughing are on both sides and not stabbing.

Pneumothorax: air in the pleural space removes the vacuum, so the lung recoils and collapses. Causes: rupture of air-filled blebs on the lung surface, or trauma. Chest tubes suck the air out so the lung re-expands.

Mechanics of breathing

Inspiration: the diaphragm and external intercostals contract, enlarging the chest top to bottom, side to side, and front to back. Pressure inside drops and air flows in. The scalenes and sternocleidomastoids help. Expiration at rest is passive; abdominal and internal intercostal muscles can add effort. A spinal cord injury that stops these muscles stops air movement.

Lung compliance: how easily the lung expands. A new balloon is hard to blow up, noncompliant. Decreased compliance: fibrosis or scar tissue, pulmonary edema because fluid is heavy, A R D S, atelectasis, and lack of surfactant. Increased compliance comes from losing elastic recoil, as in emphysema and aging: easy to inflate, hard to deflate.

Airway resistance depends on diameter. Narrow airways mean more work of breathing. Resistance is lower on inhalation, since chest expansion pulls airways open, and higher on exhalation, so constriction, like asthma, makes exhaling hard. Old air stays in, causing air-trapping, so fresh air can't get in. Children's small airways get into trouble faster. Wheezing is air through very narrow airways; wheezing on both inhalation and exhalation is more serious.

Ventilation, perfusion, diffusion

Ventilation is air moving in and out; think of a house's ventilation system. Perfusion is blood flow to the alveoli. Diffusion is gas moving across the membrane.

Only two ventilation-perfusion mismatches. Shunt: perfused but not ventilated, so blood passes from the venous to the arterial side without picking up oxygen. Causes: alveolar collapse, airway obstruction, decreased chest expansion, and a depressed respiratory center. Dead air space: ventilated but not perfused, so air reaches alveoli with no blood. The classic example is pulmonary embolism. Other causes: fat embolism from fractures, amniotic fluid embolism in childbirth, and severe vasoconstriction.

Pulmonary embolism: D V T is one of the most common causes. After a long flight, walking can knock a clot loose, so do leg exercises while seated. Findings: hypoxia, shortness of breath, rapid breathing, rapid

heart rate, possible ischemic lung damage, and inflammation causing pleuritic chest pain. Small clots may be found only at autopsy; big ones cause shock and death. Treatment: embolectomy, or for poor surgical candidates, thrombolytics then anticoagulants to prevent recurrence.

Partial pressure is the pressure one gas exerts in a mixture; total atmospheric pressure is the sum. Gases diffuse from higher to lower partial pressure, and the bigger the gradient, the more diffuses. That's why supplemental oxygen works. Diffusion also depends on surface area, about a tennis court, and membrane thickness. Less area or a thicker membrane means less diffusion.

Gas transport and affinity

Oxygen: about 1 percent dissolved, measured as P O 2. It's the only oxygen that can leave the blood. About 98 to 99 percent is bound to hemoglobin, in storage, measured as oxygen saturation, S A O 2.

Affinity is how tightly hemoglobin holds oxygen. Normally it starts releasing when P O 2 drops to about 60. Decreased affinity, releasing early, around 80: more acid, more carbon dioxide, more heat, the byproducts of metabolism. Increased affinity, holding until about 40: less acid, less carbon dioxide, less heat, saving oxygen for later.

Carbon dioxide: about 10 percent dissolved, measured as P C O 2; about 30 percent bound to hemoglobin; about 60 percent as bicarbonate. The reaction: carbon dioxide plus water makes carbonic acid, which makes bicarbonate plus a hydrogen ion. Bicarbonate moves into plasma, and the kidneys keep or dump it for acid-base balance.

Hypoxia, hypercapnia, respiratory failure

Hypoxia is not enough oxygen to the tissues. Ischemia is not enough blood, which also cuts glucose delivery and waste removal. You can have hypoxia without ischemia, like with atelectasis.

Hypoxia signs: sympathetic first, with tachycardia, pale cool clammy skin, anxiety, diaphoresis; then brain changes, unclear thinking, restlessness, impaired judgment, confusion, coma. Cyanosis is late. Capillary refill checks circulation, not oxygen.

Hypercapnia, high carbon dioxide, causes C O 2 narcosis: progressively sleepy, then disoriented, then coma, with respiratory depression. Hypoxia and hypercapnia usually occur together.

Respiratory failure: P O 2 below 50 and or P C O 2 above 50. Normal P O 2 is 80 to 100; normal P C O 2 is 35 to 45. It's a condition any serious respiratory disease can cause, not a disease, and not respiratory arrest; the patient is still breathing. These numbers are for acute cases; chronic patients can sit there alert, so look at the patient, not just the numbers.

Control of breathing

The respiratory center is in the pons and medulla; very irregular breathing is a sign of brainstem injury. Chemoreceptors in the medulla, carotids, and aorta track oxygen, carbon dioxide, and pH. Lung receptors track pressure, stretch, congestion, and irritation. The cortex allows voluntary override. The autonomic nervous system affects only airway diameter: sympathetic causes bronchodilation, because to fight or run you need oxygen; parasympathetic causes bronchoconstriction.

Rapid recap

S A 60 to 100, A V 40 to 60, Purkinje 15 to 40. P wave is atria, Q R S is ventricles depolarizing, T is ventricles repolarizing. Normal sinus means a P before every Q R S and even R-R intervals. Dysrhythmias matter because they drop cardiac output. Causes: congenital, damage, inflammation, electrolytes, drugs. A fib means lost kick and clots. V fib means zero output, defibrillate. Fix the cause first. Type one cells exchange gas; type two make surfactant; no surfactant, atelectasis. Shunt means perfused not ventilated; dead space means ventilated not perfused, like P E. More acid, carbon dioxide, and heat make hemoglobin let go. Respiratory failure is P O 2 under 50 and P C O 2 over 50. Sympathetic dilates airways; parasympathetic constricts.

Class 9. Respiratory Function, Part 2

Respiratory defenses

Four defenses. The mucociliary blanket removes secretions, microorganisms, and particles from the respiratory tract. Pulmonary macrophages remove organisms and foreign particles from the lung. The glottic and cough reflexes protect against aspiration into the tracheobronchial tree. Mucosa-associated lymphoid tissue traps and processes antigens and brings them to mature immune cells. The pathological change in this class is inflammation of respiratory tissues. Predisposing factors: impaired immunity, impaired defenses, smoking and irritating gases, and exposure to organisms. The main nursing problem is impaired oxygenation.

Pneumonia

Pneumonia is inflammation of the respiratory lung tissue, where the alveoli and pulmonary capillaries are and gas exchange happens. Not the conducting airways. It can be caused by organisms, irritating fumes, or aspirated gastric acid; here we focus on infection. Symptoms depend on which structures are involved and how severe it is.

Typical pneumonia is usually bacterial. The alveolar walls inflame, capillaries leak, and exudate and mucus fill the alveoli. Result: less surface area, a thicker alveolar-capillary membrane, and heavier, less compliant lungs, so diffusion is harder and the work of breathing rises. These patients are usually sicker.

Atypical pneumonia is usually viral, also called interstitial. Inflammation is in the air ducts and interstitial spaces, not inside the alveoli. Less sputum, milder symptoms, "walking pneumonia." The worry is a secondary bacterial pneumonia on top.

Because of overlap, the preferred classification now is community-acquired versus hospital-acquired or health care-associated. Hospital organisms are more virulent and more likely to have been exposed to antibiotics, so they're resistant and harder to treat.

Lobar pneumonia involves a whole lobe. Bronchopneumonia is patchy, scattered across several lobes.

Pneumonia predisposing factors

Impaired immunity: aging, immunosuppressive drugs, AIDS. Pre-existing lung disease. Chronic disease of other systems, causing weakness and less mobility. Immobility makes deep breathing and coughing hard, so secretions collect and bacteria grow; pneumonia is a preventable complication of immobility. I C U and ventilator patients, since organisms grow in tubing. Poor oral care, because mouth organisms get aspirated; brushing teeth is a health measure, not just comfort. Exposure to organisms, at home or work.

Impaired lung defenses: a weak cough reflex, as in spinal cord injury where abdominal muscles can't power a cough; an impaired mucociliary blanket, since dehydration thickens mucus; and loss of alveolar macrophages. Smoking kills macrophages, paralyzes cilia, and irritates airways.

Pneumonia findings and treatment

Classic signs: fever, usually 101 to 102; chills; malaise, meaning tired, lousy, body aches; a productive cough with thick yellow, green, brown, or rust-colored sputum, where rust means blood; crackles;

shortness of breath; and pleuritic chest pain, sharp and stabbing with coughing or deep breaths, if inflammation reaches the pleura. Sputum color isn't reliable for identifying the organism.

Elderly patients often lack classic signs. They may not mount a fever or have the energy to cough. Often the only sign is a decreased level of consciousness because the brain isn't getting enough oxygen. Test hook: nursing-home patient with sudden confusion, think pneumonia.

Treatment: antibiotics and oxygen. Antibiotics treat bacterial pneumonia. They don't kill viruses but may be given to high-risk, especially immunosuppressed, viral pneumonia patients to prevent secondary bacterial infection. Antivirals aren't routine because of side effects; milder ones like Tamiflu if needed. Oxygen is a drug and can be toxic, so check the blood P O 2 first and use it only if needed.

Tuberculosis basics

T B is caused by Mycobacterium tuberculosis, an acid-fast bacillus, A F B. A sputum culture for A F B means you're looking for T B.

Transmission is airborne through aerosolized secretions from someone with active T B. It requires close or prolonged contact; a brief exposure is unlikely to infect. It's not spread by doorknobs, glasses, or toilet seats. Risk rises with more organisms in the air, from frequent coughing without covering, and with weaker host immunity. M D R, multi-drug resistant, and X D R, extensively drug-resistant, strains exist, so wear an N 95 mask with active T B.

The organism is strictly aerobic, so it prefers high-oxygen areas. The lungs are most common, but extrapulmonary T B can affect the meninges, kidneys, and adrenal glands, so assess other systems.

Latent versus active T B

After exposure, the organism implants in an alveolus or airway. Macrophages engulf it but can't kill it because of its waxy capsule. So the body walls it off in a granuloma, also called a tubercle or Ghon focus. Scar tissue forms, may calcify, and the organisms go dormant.

Latent T B infection, L T B I: infected, with a positive skin or blood test, but no symptoms, not contagious, normal chest X-ray, negative sputum culture. Most people take this path.

Reactivation: live organisms stay in the tubercle. If the person later becomes immunosuppressed, by chemotherapy or H I V, they reactivate and multiply into active T B. Someone already immunosuppressed when first infected can go straight to active disease.

Active disease: the tubercle's center necroses, cheesy, enlarges, and erodes into an airway. Necrotic material with live organisms gets coughed up, so the person is contagious, and it leaves a cavity. Two bad results: contagious, and less surface area for gas exchange. Early cavities are tiny and may not show on X-ray; later they coalesce. In active disease the skin or blood test is positive, symptoms are present, and sputum culture is positive. The X-ray may or may not be abnormal, so a normal X-ray doesn't rule it out.

T B testing

The P P D, purified protein derivative, is the antigen; Mantoux is the intradermal method. It needs two visits, read in two to three days. It only tells you someone is infected, not whether it's active.

The blood test is the I G R A, interferon gamma release assay; brands are QuantiFERON Gold and T-SPOT. Blood is mixed with a peptide mimicking T B antigen; if white cells release interferon gamma, it's positive. It also can't tell active from latent. Advantages: one visit, results in about 24 hours, and fewer false positives after the B C G vaccine. Downsides: a blood draw, harder in kids, lab processing, paperwork, and cost. The skin test is more cost-effective for screening large groups.

The chest X-ray can't tell if disease is active. The gold standard for active T B is a sputum culture for A F B. Organisms in sputum mean active and contagious.

High-risk groups: H I V or immunosuppressed; people born in or who traveled to high-T B countries; crowded settings like prisons, drug treatment facilities, and homeless shelters; poverty with overcrowding, poor ventilation, poor nutrition and hygiene; health care workers; and children under five.

Active T B findings and treatment

Systemic signs come first and are nonspecific: fatigue, anorexia and weight loss, low-grade fever under 100, about 99.3 to 99.5, and night sweats. Low-grade fever and night sweats are the classic T B signs. Respiratory signs come late: a dry cough becoming productive, sounding like pneumonia, and dyspnea. T B is often treated as pneumonia first and suspected when treatment fails, so take a good history and follow up red flags in high-risk groups.

Anti-T B drugs are used almost only for T B and have more side effects, so monitor closely. You don't need drug names. Usually three or four drugs with different mechanisms, because T B mutates and becomes resistant fast. The course is long, six to nine months, because tubercles have poor blood supply and organisms go dormant. Not finishing breeds resistance, so health departments use directly observed therapy, in person or by video. After about two weeks on meds, patients are no longer contagious, so no isolation or sanitarium.

Latent T B: chemoprophylaxis with one drug, usually I N H, isoniazid, for about a year. It's the patient's choice.

Restrictive versus obstructive

Restrictive disorders mean trouble expanding the lungs, from congestion or fibrosis. Obstructive disorders mean limited airflow on expiration, usually from narrowed airways: decreased elastic recoil, inflammation, more mucus. Asthma and C O P D are obstructive.

Asthma

Also called reactive airway disease, R A D, or reversible obstructive airway disease, R O A D. It's a type one, I g E-mediated hypersensitivity: genetic predisposition, a family history of allergies, plus a trigger, with airway hyperresponsiveness. It's chronic with acute episodes. It's reversible, but long-term inflammation can cause permanent damage, so treat even mild asthma.

Early phase, 10 to 20 minutes after the trigger: antigen binds I g E on mast cells, which release mediators, causing bronchospasm, mucosal edema, and more mucus. It resolves in about 60 to 90 minutes. Late phase, 2 to 8 hours later: white cells recruited by those mediators release more mediators, making the airways more inflamed and reactive. A vicious cycle that can damage epithelium.

Triggers: allergens like pollen, dust, and dander; respiratory infection, which adds inflammation; cold air and exercise, because fast, deep breathing loses heat and moisture from the airways, and cold air is drier; gastroesophageal reflux with aspirated acid; emotional upsets; drugs and chemicals; pollutants and workplace fumes; second-hand smoke; and hormonal shifts like puberty, pregnancy, and menopause.

Findings: wheezing, air through very narrow airways; hypoxia with shortness of breath and fast breathing; chest tightness with a repeated dry cough, since coughing briefly stretches the airways open, becoming productive as mucus increases; prolonged expiration, because airways are narrower on exhalation; and more secretions. Count respirations when the patient doesn't know. A severe attack can cause respiratory failure.

Interventions: identify and avoid triggers, since preventing an attack is easier than stopping one. Bronchodilators. Drugs blocking inflammatory mediators, like leukotriene modifiers such as Singulair, which work better than antihistamines; histamine isn't a major driver, and antihistamines dry secretions.

C O P D

Chronic obstructive pulmonary disease: chronic, non-reversible airway obstruction and inflammation. Most common cause is smoking; also second-hand smoke, workplace fumes, and alpha-one antitrypsin deficiency. Alpha-one antitrypsin is an enzyme that protects the lungs from inflammatory damage. Deficiency can be inherited, autosomal recessive, or come from smoking or repeated infections. Two forms: emphysema and chronic bronchitis. Many patients have both, so the diagnosis is often just C O P D.

Emphysema, type A, the pink puffer

Two events. One, inflammatory enzymes destroy alveolar walls and their capillary beds, so air spaces become fewer and bigger, with less surface area, risking hypoxia and hypercapnia. Two, loss of elastic recoil makes it hard to push air out, so air is trapped and the lungs hyperinflate. Old air can't get out, so fresh air can't get in.

Chronic air trapping causes barrel chest: the front-to-back, or anteroposterior, A P, chest dimension nearly equals side to side. Document it as increased A P chest dimensions.

Pink puffer: they work hard to breathe, puffing, but keep blood gases fairly normal until late, so they stay pink. Shortness of breath, weight loss from the energy cost of breathing and trouble eating while dyspneic, though the barrel chest can hide it, and decreased breath sounds from less air movement.

Chronic bronchitis, type B, the blue bloater

Don't confuse it with acute bronchitis, usually a viral head cold that moved to the chest. Chronic bronchitis is not an infection; it's inflammation from smoking or fumes. Emphysema destroys alveolar walls; chronic bronchitis affects the bronchi, the conducting airways.

Two changes. One, chronic inflammation causes mucosal edema, then fibrosis, permanently narrowing the airway. Two, mucus cells increase in number and size, making more thick, sticky mucus that's hard to cough up and further obstructs the airway, especially on expiration. That mucus is a protective reaction that makes oxygenation worse.

Blue bloater: they work hard but can't keep blood gases normal, so they become hypoxic and hypercapnic more easily and look blue. Findings: wheezing and crackles, prolonged expiration, shortness of breath, poor activity tolerance, and a chronic productive smoker's cough with thick yellow sputum that looks like pneumonia but isn't an infection.

Some develop cor pulmonale, right heart failure from lung disease. Normally, vessels in poorly ventilated areas constrict to send blood to well-ventilated areas. When nearly the whole lung is poorly ventilated, chronic hypoxia constricts the vessels widely, causing pulmonary hypertension. The right ventricle overworks and fails. Blood backs up into the veins: J V D, peripheral edema, engorged abdominal organs. The edema is the "bloater."

C O P D interventions

Avoid what worsens inflammation: smoking and respiratory infections. An infection on top of C O P D doubles the inflammation and may mean hospitalization. Since C O P D isn't reversible, the goal is to return to baseline. Bronchodilators, with limited effect once fibrosis is permanent.

Oxygen, carefully. Goals: relieve shortness of breath and treat hypoxia, which also helps prevent cor pulmonale. Check blood gases first. Normally, low oxygen or high carbon dioxide drives breathing. In long-term C O P D, the carbon dioxide drive is blunted, so low oxygen becomes the main drive. Too much oxygen, a P O 2 above about 60, can remove that drive and cause respiratory depression, a rate of 6 to 8 when normal is 12 to 20. So use low-flow oxygen, no more than 2 liters per minute by nasal cannula, or titrate to keep P O 2 between 55 and 65. Example: start at 1 liter; if P O 2 is about 45, go to 2. More than 2 liters isn't always wrong, like when P O 2 is still 40 on 2 liters, but it's a red flag: check the rate and oxygenation.

Breathing mechanics review

Air moves down a pressure gradient created by chest size changes. Inspiration: muscles enlarge the chest. Expiration: mostly elastic recoil. Compliance decreases with scar tissue, congestion, or less surfactant, and increases when elastic recoil is lost. Airway resistance depends on diameter; bronchospasm and secretions increase it and the work of breathing. Resistance is lower on inspiration and higher on expiration.

Rapid recap

Pneumonia inflames the gas-exchange tissue; typical is bacterial, exudate in alveoli; atypical is viral and milder. Hospital-acquired is more resistant. Immobility and poor oral care set it up. Elderly may only be confused. Treat with antibiotics and oxygen after checking P O 2. T B: airborne, prolonged contact, waxy capsule, granuloma. Latent means positive test, no symptoms, not contagious; active means cavities, contagious, positive sputum culture, the gold standard. P P D and I G R A can't tell latent from active. Four drugs, six to nine months, not contagious after two weeks; I N H for latent. Asthma is I g E-mediated and reversible, with early and late phases. Emphysema destroys alveoli, traps air, barrel chest, pink puffer. Chronic bronchitis means fibrosis and mucus, blue bloater, cor pulmonale. C O P D oxygen: no more than 2 liters, P O 2 55 to 65.

Class 10. Fluid, Electrolyte, and Acid-Base Balance

Why it matters and test scope

Fluid, electrolyte, and acid-base imbalances aren't diseases themselves; they go along with many diseases, like fever. But they can be incompatible with life. Test scope: the normal sources of gain and loss and the risk factors for each imbalance are in the study guide, and she'll test them as patient scenarios, like "this patient has hyperkalemia; why?" Link them: a normal route in that fails, like vomiting so you can't drink, or a route out that fails, like kidney failure so you can't make urine, is the risk factor.

Compartments

Intracellular fluid is inside cells, about two-thirds of body water. Extracellular fluid includes intravascular, in vessels; interstitial, between cells; lymph; and transcellular fluid, in body spaces like the peritoneal, pleural, pericardial, and joint spaces, cerebrospinal fluid, and digestive juices, normally small amounts. Third-spacing is an abnormally large amount collecting in one of those spaces, where it can't be used, like a pericardial effusion, ascites, peritonitis, pancreatitis, intestinal obstruction, or burn edema.

Tonicity

Water between cells and interstitium depends on the tonicity, or osmolality, of extracellular fluid. Hypertonic extracellular fluid pulls water out of cells, so they shrink. Hypotonic lets water into cells, so they swell. Isotonic causes no change.

Capillary forces and edema

Four forces move fluid between vessels and interstitium. Capillary hydrostatic pressure pushes fluid out of the capillary; it's a mechanical force set by the volume of fluid inside. Capillary colloidal osmotic pressure pulls fluid into the capillary; colloidal means protein, mainly albumin, which is too big to leave. Tissue hydrostatic pressure pushes fluid from the tissues back into the capillary. Tissue colloidal osmotic pressure, from tissue proteins, pulls fluid into the interstitium. Normally a small amount of extra fluid is left in the tissues, and the lymph system returns it. Change any one force and you get edema.

Examples. Decreased capillary colloidal osmotic pressure: malnutrition lowers albumin, so fluid leaks out, like the swollen bellies of starving children. Increased capillary hydrostatic pressure: a clot in a leg vein pools blood and pushes fluid out. Inflammation makes capillaries leaky, proteins escape, tissue colloidal osmotic pressure rises, and fluid is pulled into tissues. Impaired lymph flow causes lymphedema: after a radical mastectomy removes axillary lymph vessels, the arm swells, so no blood pressures, venipunctures, or I Vs in that arm. Fluid problems can be about where the fluid is, not just how much.

Water gains, losses, regulation

Functions of body water: maintain vascular volume and provide the environment for cell metabolism. Gains: oral intake, I V fluids, tube feedings, and oxidation of nutrients, metabolic water, about 300 milliliters a day. Losses: urine, insensible loss through skin and lungs, stool, sweat, which rises with heat, fever, and exercise, and abnormal losses like vomiting, drains, ostomies, burns, and bleeding. Normal intake roughly equals output, 2000 to 2500 milliliters a day. Regulation: thirst controls intake; A D H increases water reabsorption in the collecting duct.

Isotonic fluid volume deficit: hypovolemia

Also called volume contraction or depletion: a proportionate loss of sodium and water from the extracellular compartment. Only volume changes, not concentration. Losing water without sodium would concentrate the fluid. Hemorrhage is isotonic.

Risk factors. Impaired intake: decreased mobility, coma, no access, impaired thirst or swallowing, nausea, anorexia, N P O. Excess losses: G I, vomiting, diarrhea, suction, ostomy or fistula drainage; renal, polyuria, diuretics, osmotic diuresis from uncontrolled diabetes, adrenal insufficiency with low aldosterone; skin, sweating from fever or exercise, burns; and third-spacing.

Findings: low blood pressure. Severe means low all the time, up to hypovolemic shock; milder means orthostatic, or postural, hypotension, normal lying down and dropping on standing, so check vitals lying, sitting, and standing. Heart rate rises to compensate. Thready, or thin, pulses describe quality, not rate. Dry mucous membranes. Decreased skin turgor, useful only in younger people. Concentrated urine and decreased output. Thirst. Weight loss, with a pound equaling about 500 milliliters. Temperature goes up, because less blood carries heat to the skin.

Interventions: replace fluid, orally with an isotonic electrolyte solution if they can drink, I V if not. And find and fix the cause.

Isotonic fluid volume excess: hypervolemia

Also called fluid expansion or overload: a proportionate gain of sodium and water. Volume only.

Risk factors. Decreased elimination: renal failure; heart failure, from decreased renal blood flow; excess corticosteroids; cirrhosis, with low albumin and high aldosterone and A D H; nephrotic syndrome. Excess intake: high-sodium diet, processed and canned foods; fast or excessive I V normal saline, lactated Ringer's, or blood products.

Findings: high blood pressure; full, bounding pulses, again quality not rate; jugular venous distention; peripheral and dependent edema, at the ankles or the sacrum in bed; weight gain; and crackles if there's pulmonary edema.

Interventions: mild, restrict fluid and sodium, since sodium holds water. Severe, with crackles or oxygenation problems, diuretics.

Electrolytes

An electrolyte develops an ionic charge in water. Glucose doesn't, so it's not one. Cations are positive; anions are negative. Uses: fluid balance, p H balance, and action potentials in nerve and muscle, like spark plugs. Focus on sodium, potassium, and calcium. Know each one's function, then predict what too little or too much does.

Sodium

The main extracellular cation, normal 135 to 148. The sodium-potassium pump moves sodium out and potassium in. Sodium controls extracellular osmolality, so it decides whether cells shrink or swell, and it regulates vascular volume. It also helps with resting membrane potential and action potentials. Gains: diet, plus I V fluids and meds containing sodium, like normal saline, sodium bicarbonate, some I V antibiotics,

and tube feedings. Losses: kidneys, a little in stool, sweat, and abnormal losses like vomiting, diarrhea, and drains.

Hyponatremia, under 135, makes extracellular fluid hypotonic, so cells, including brain cells, swell. Neuro checks are the priority. Two forms. Depletional: sodium lost, same water. Dilutional: same sodium, extra water. Dilutional is more common, also called water intoxication. Example: endurance athletes lose sodium and water in sweat but replace only plain water. A radio contest to drink the most water without urinating killed the winner.

Hyponatremia risk factors: sodium and water losses replaced with sodium-free fluid, from sweating, burns, G I losses, or aggressive diuretics; sodium-free fluid means plain water, D 5 W, or ice chips with N G suction; tap water enemas; excess water intake, including psychogenic polydipsia; adrenal insufficiency, Addison's, where low aldosterone dumps sodium; increased A D H, S I A D H, from lung cancer, head injury, pain and stress after surgery, or some drugs; heart failure and cirrhosis; and too much hypotonic I V fluid like D 5 W or half-normal saline.

Neuro signs by severity: headache and lethargy; nausea and motor weakness; then confusion, seizures, coma, death.

Interventions: depletional, give sodium, I V normal saline. Dilutional, restrict water. Both, or neuro symptoms, hypertonic saline plus a diuretic. Always fix the cause.

Hypernatremia, over 148: extracellular fluid is hypertonic, so cells shrink, intracellular dehydration. Causes: excess sodium, like near-drowning in seawater, hypertonic saline, I V sodium bicarbonate, or concentrated tube feedings without enough free water; or water loss without sodium, like A D H deficiency, polyuria, osmotic diuresis from high blood sugar or mannitol, watery diarrhea, sweating, fever, burns, a hot environment; or decreased intake, in infants, older adults, and unconscious or N P O patients.

Signs: dry skin, dry mucous membranes, no tears, like a crying newborn with no tears, and neuro signs: headache, seizures, coma. If from water loss, also weight loss, thirst, and higher temperature. Do neuro checks.

Interventions: excess sodium, restrict sodium. Water loss, replace water.

Potassium

The main intracellular cation, normal 3.5 to 5.0. It regulates intracellular osmolality, maintains resting membrane potential and action potentials, and affects acid-base through the plasma potassium-hydrogen exchange. Gains: fruits and vegetables like bananas, potatoes, oranges, tomatoes; meats and dairy; supplements; salt substitutes, which are potassium chloride; I V potassium. Losses: mainly the kidneys, so watch kidney function and urine output, plus stool and sweat. Aldosterone makes the kidney save sodium and dump potassium.

Plasma potassium-hydrogen exchange: if blood potassium is low, potassium moves out of cells and hydrogen moves in. If there's too much acid, hydrogen moves into cells and potassium moves out. Potassium changes p H, and p H moves potassium.

Hypokalemia, under 3.5: more potassium leaks out of cells, so it's harder to generate an action potential; more sodium is needed to depolarize. Hydrogen moves in, so there's some alkalosis. Skeletal muscle: weakness, fatigue, cramps. Heart: dysrhythmias, decreased cardiac output, blood pressure changes; the

heart is the highest priority. G I smooth muscle: decreased bowel sounds and peristalsis, nausea, distension, constipation. Kidneys try to save potassium by dumping sodium, water follows, causing polyuria and thirst.

Risk factors: inadequate intake, like long N P O or potassium-free I V fluids, anorexia or eating disorders, alcoholism, malnutrition; excess loss, like loop and thiazide diuretics, high aldosterone or glucocorticoids, low magnesium, N G suction, vomiting, laxative abuse, ostomy drainage, sweating, burns; and shifts into cells, from alkalosis, insulin, beta agonists like albuterol, and refeeding after starvation.

Treatment: replace potassium, orally if they can swallow, otherwise in the I V bag over about 8 to 10 hours.

Hyperkalemia. Mild: less potassium moves out, so action potentials are easier: numbness and tingling of the lips or fingers, increased peristalsis, mild diarrhea; some notice nothing. Severe: potassium can't move out, cells can't repolarize, no action potentials: skeletal muscle paralysis, no peristalsis, dysrhythmias, cardiac arrest. Never push potassium I V; always add it to a bag and infuse slowly.

Risk factors: decreased elimination, renal failure, aldosterone deficiency, adrenal insufficiency, potassium-sparing diuretics, A C E inhibitors, A R Bs, N S A I Ds; excess intake, supplements, salt substitutes, I V potassium too fast, stored old blood; shifts out of cells, from burns, crush injuries, cell damage, extreme exercise, acidosis, rhabdomyolysis, tumor lysis, hemolysis, and insulin deficiency in uncontrolled diabetes.

Treatment: mild, low-potassium diet. Higher, drugs that excrete potassium through urine or G I tract, which take several hours; you don't need the drug names. Dysrhythmias present, dialysis, which removes potassium in a few hours. Check muscle function, especially the heart.

Calcium

Calcium is a divalent cation. Most is in bone and teeth; of the rest, most is intracellular. The tiny extracellular amount causes the imbalances. Normal total is 8.5 to 10.5. About half of serum calcium is bound to albumin and stuck in the blood; only free, ionized calcium participates in cell function. Calcium affects membrane potential and permeability and is needed for contraction of all muscle types. Calcium and phosphorus are reciprocal, and calcium and magnesium affect each other.

Gains: dairy, green leafy vegetables, fortified foods, supplements, I V calcium, with vitamin D needed for absorption. Losses: mostly urine, plus stool and sweat. Regulated mostly through the kidneys. P T H, parathyroid hormone, maintains serum calcium: it increases renal reabsorption, enhances vitamin D activation by the kidneys, and releases calcium from bone. Vitamin D maintains bone mineralization: it increases intestinal absorption and renal reabsorption and decreases P T H.

Calcium is the membrane stabilizer: when present, it's harder for sodium to enter, so action potentials are harder. When absent, they're easier.

Hypocalcemia, under 8.5: easier action potentials. Paresthesias, numbness and tingling; tetany, twitching and spasms of the face, hands, and feet; laryngospasm, which narrows the airway with stridor, a harsh high-pitched sound, an emergency; seizures; and dysrhythmias. Mnemonic CATS: convulsions, arrhythmias, tetany, spasms and stridor. Cats aren't hyper, so CATS is hypocalcemia. But rely on the logic.

Risk factors: decreased intestinal absorption, from low vitamin D, malabsorption, low dietary calcium, and chronic kidney disease, because kidneys can't activate vitamin D; decreased mobilization from bone, like hypoparathyroidism after neck or thyroid surgery, and low magnesium, which lowers P T H; abnormal renal loss, like loop diuretics; high phosphate, as in kidney failure; and calcium trapped or bound, in acute pancreatitis, massive transfusion of citrated blood, and alkalosis.

Treatment: acute, I V calcium. Chronic hypocalcemia weakens bones, causing osteoporosis, fractures, and deformities; treat with oral calcium plus vitamin D.

Hypercalcemia, over 10.5: harder action potentials. Muscle weakness and flaccid muscles, limp with no tone or reflexes; constipation, decreased bowel sounds, nausea and vomiting; stupor, eyes open but not responding; dysrhythmias. Chronic: kidney stones, mostly calcium-based.

Risk factors: increased bone resorption, from excess P T H, hyperparathyroidism, immobility, bone disorders, bone cancers, metastases, multiple myeloma; increased intestinal absorption, from excess vitamin D or dietary calcium, supplements or calcium antacids; decreased renal excretion, from thiazide diuretics or dehydration.

Treatment: fix the cause, like parathyroid or bone disease; low-calcium diet; drugs that excrete calcium; and osteoporosis drugs that keep calcium in bone. No drug names needed.

Alkalosis and calcium: calcium and hydrogen compete for binding sites on albumin. In alkalosis, there's less hydrogen, so more calcium binds albumin, and free calcium drops. That's why alkalosis looks like hypocalcemia, with tetany, dizziness, and paresthesias, even with normal total calcium.

Acid-base basics

Normal p H is 7.35 to 7.45, needed for membrane excitability, enzyme function, and protein structure. p H depends on relative hydrogen ion concentration: more hydrogen, more acid, lower p H. An acid releases hydrogen; a base accepts it.

Metabolism constantly makes acid. Only two routes remove it. The lungs excrete the volatile acid, carbonic acid, as carbon dioxide, within minutes, but can't fully normalize p H alone. The kidneys excrete all the other, nonvolatile acids; they take hours to start but keep going for days until p H is normal. Meanwhile, buffers and compartmental shifting prevent big swings.

The bicarbonate buffer system is the most important extracellular buffer. Carbon dioxide plus water makes carbonic acid, which makes hydrogen plus bicarbonate. Carbonic acid is the acid part, bicarbonate the base part. With too much acid, bicarbonate picks up hydrogen and makes carbonic acid, a weak acid that breaks into carbon dioxide and water, which is exhaled: strong acid becomes weak acid. With too much base, carbonic acid releases hydrogen, and the leftover bicarbonate is a weak base the kidneys can excrete. Other buffers: protein, phosphate, and bone. Compartmental shifting is the plasma potassium-hydrogen exchange, moving extra hydrogen into cells to buy time.

A B G values and the 20 to 1 ratio

Memorize these; they're not given on the test, and expect two or three A B Gs. P a C O 2, 35 to 45, the respiratory indicator; carbon dioxide is acid. The "a" means arterial. H C O 3, bicarbonate, 22 to 26, the metabolic indicator; it's base. p H, 7.35 to 7.45.

For a normal p H, bicarbonate to carbonic acid must be 20 to 1. That's carbonic acid, not carbon dioxide, so the normal numbers won't look like 20 to 1. p H depends on the ratio, like a scale. If the lungs retain carbonic acid, the kidneys can retain bicarbonate to keep the ratio. That's compensation.

Interpreting A B Gs

Step one: p H. Acidosis or alkalosis? Step two: P a C O 2 and bicarbonate, high, low, or normal. High C O 2 is more acid; low is more alkaline. High bicarbonate is more alkaline; low is more acid. Step three: which one caused it? The cause pushes p H the same direction as the p H change. C O 2 means respiratory; bicarbonate means metabolic. Step four: is the other value compensating? Uncompensated: the other value is normal. Partially compensated: the other value is moving the right way, but p H is still abnormal. Fully compensated: p H is normal, but both values are abnormal; find the original problem by which end of normal the p H is closer to, 7.36 means acidosis, then pick the value that matches. You only need the numbers, not the story; P a O 2 isn't needed to name the imbalance, but still check oxygenation.

Respiratory imbalances are carbon dioxide problems from changed ventilation or gas exchange; everything else is metabolic. The kidneys compensate for respiratory problems by changing urine p H. The lungs compensate for metabolic problems by changing rate and depth: faster and deeper to blow off acid, slower and shallower to keep it.

Findings in any acid-base disorder come from three things: the p H itself, the compensation, and the underlying disease. Acidosis decreases membrane excitability. Alkalosis increases it.

The four imbalances

Metabolic acidosis: primary base deficit; low p H, low bicarbonate. Lungs compensate by breathing faster and deeper. Risk factors: excess metabolic acids, including lactic acidosis from shock, sepsis, hypoxia, cardiac arrest, or severe exercise; ketoacidosis from uncontrolled diabetes, D K A, starvation, or heavy alcohol use; renal failure; and toxins like aspirin, ethylene glycol, and methanol. Loss of bicarbonate: diarrhea, drainage tubes, fistulas, ileostomy output, renal tubular acidosis, carbonic anhydrase inhibitors. Hyperchloremia, as from large volumes of normal saline: chloride and bicarbonate are both anions, so high chloride means low bicarbonate. Symptoms of decreased excitability: lethargy, confusion, stupor, coma; slower heart rate, weaker contraction, lower cardiac output, dysrhythmias. Assess excitability, compensation, and the cause. Treat the cause, then reassess whether it's getting better.

Metabolic alkalosis: primary base excess; high p H, high bicarbonate. Lungs compensate by breathing slower and shallower, which only goes so far before hypoxia and hypercapnia, so there's a risk of respiratory failure. Risk factors: excess bicarbonate or alkali, from sodium bicarbonate, antacid overuse, or lactate, citrate, or acetate in I V fluids or massive transfusion; loss of hydrogen, from vomiting or gastric suction, which lose hydrochloric acid and chloride, and low chloride means high bicarbonate; diuretics, which lower potassium, so kidneys dump hydrogen to save potassium and hydrogen shifts into cells; and excess aldosterone or steroids, Cushing's. Symptoms of increased excitability: paresthesias, tetany, confusion, and if severe, seizures and dysrhythmias. Fix the cause.

Respiratory acidosis: primary carbonic acid excess; low p H, high P a C O 2. Kidneys compensate by excreting more acid and keeping base, but that takes a day or more, so p H can fall a lot first. Cause: impaired alveolar ventilation. Respiratory center depression from head injury, overdose, opioids, sedatives, anesthesia, or stroke. Lung disease like C O P D, pneumonia, A R D S, pulmonary edema, severe asthma, atelectasis. Airway obstruction from aspirated food, a mucus plug, bronchospasm, or the

tongue in an unconscious patient. Chest problems like flail chest, pneumothorax, painful rib fractures, obesity hypoventilation. Respiratory muscle disorders like Guillain-Barré, myasthenia gravis, A L S, spinal cord injury, or fatigue. Symptoms overlap with hypoxia and hypercapnia: headache, confusion, stupor, coma, C O 2 narcosis. Treat by improving ventilation and gas exchange.

Respiratory alkalosis: primary carbonic acid deficit; high p H, low P a C O 2. Kidneys compensate by excreting less acid and more base, though episodes are often too short. Cause: hyperventilation, breathing faster than needed, from anxiety or panic, severe pain, fever, hypoxia and lung disease like high altitude, pulmonary embolism, sepsis, early aspirin overdose, stroke or head injury, or a ventilator rate set too high. Symptoms: numbness and tingling around the mouth, dizziness, tetany, seizures. Fix the cause; breathing into a paper bag rebreathes carbon dioxide.

Her worked examples

Baby L, with an N G tube to suction after bowel resection: p H 7.49, C O 2 52, bicarbonate 33. Alkalosis, caused by high bicarbonate, so metabolic alkalosis; C O 2 is high, compensating, but p H is still abnormal: partially compensated. Suction removed gastric acid.

Mr. B, pneumonia: p H 7.32, C O 2 55, bicarbonate 26. Acidosis from high C O 2, respiratory; bicarbonate normal: uncompensated respiratory acidosis.

Ms. C, anxiety: p H 7.58, C O 2 28, bicarbonate 18. Alkalosis from low C O 2; low bicarbonate is trying: partially compensated respiratory alkalosis.

Mr. E, coma: p H 7.36, C O 2 50, bicarbonate 29. p H normal but closer to acid; high C O 2 matches: fully compensated respiratory acidosis.

There's an A B G handout in the module and abg dot ninja for practice.

Rapid recap

Hydrostatic pressure pushes; colloidal osmotic pressure, from proteins, pulls. Isotonic deficit and excess are volume problems: deficit gives orthostatic hypotension and thready pulses; excess gives bounding pulses, J V D, edema, crackles. Sodium problems are brain problems: low means cells swell; high means cells shrink; do neuro checks. Low potassium makes action potentials harder, so weakness and dysrhythmias; high potassium is mild tingling, then paralysis and arrest; never push potassium. Calcium stabilizes membranes: low is CATS; high is flaccid, stupor, stones. Alkalosis lowers free calcium. p H 7.35 to 7.45, C O 2 35 to 45, bicarbonate 22 to 26. p H first, find the cause, check compensation. Acidosis depresses; alkalosis excites.

Class 11. Alterations in Renal Function

Kidney functions: A WET BED

The kidney filters blood, saves what you need, gets rid of what you don't, and for things you need only some of, like fluid and electrolytes, keeps what you need and dumps the excess.

Excretory functions: excrete metabolic wastes, regulate fluid volume and so blood pressure, regulate electrolytes, and regulate acid-base balance. Wastes: urea, from protein breakdown; uric acid, from nucleic acids; creatinine, from creatine phosphate, the energy compound in skeletal muscle. Nurse point: kidneys also excrete drug breakdown products, or they build to toxic levels.

Endocrine functions, from hormone-like substances: renin, which raises blood volume and pressure; erythropoietin; and vitamin D activation, with the liver.

Renin review: poor renal blood flow triggers renin, which makes angiotensin one, converted in the lungs to angiotensin two. Angiotensin two tells the adrenal cortex to make aldosterone, which retains sodium and loses potassium, with water following sodium, and it's a powerful vasoconstrictor. So the kidneys raise pressure through volume and vasoconstriction.

Erythropoietin: with hypoxia, kidneys make erythropoietin, telling the bone marrow to make red cells.

Vitamin D activation takes two steps, first the liver, then the kidneys. Sources are skin and food. Activated vitamin D is needed to absorb calcium.

Mnemonic: A WET BED. Acid-base, water, electrolytes, toxins or wastes, blood pressure, erythropoietin, vitamin D. If the kidneys fail, nearly every system is affected.

Nephron: vascular component and G F R

The nephron is the functional unit, more than a million per kidney. Two components: vascular, where blood is filtered, and tubular, which collects filtrate that becomes urine. Glomeruli and parts of the tubules sit in the outer cortex; the inner medulla holds mostly tubules, and the loops point toward it. The renal calyces and renal pelvis collect urine from the tubules and funnel it to the ureter.

Blood path: aorta, renal artery, smaller arteries, the afferent arteriole, then the glomerulus, a specialized tuft of capillaries where blood is filtered. Filtrate enters Bowman's space inside Bowman's capsule, then the tubules.

Glomerular filtration rate, G F R, is the total filtrate from all glomeruli per minute, about 125 milliliters.

The glomerular wall has three layers. The middle one, the glomerular basement membrane, is the filter; the spaces between its fibers decide what passes. Red cells and plasma proteins like albumin are normally too big, so they're normally not in urine.

Unfiltered blood leaves through the efferent arteriole. Afferent is the intake valve; efferent is the output valve. Their constriction and dilation control glomerular pressure. Afferent constricted plus efferent dilated lowers pressure and G F R. Afferent dilated plus efferent constricted raises pressure and G F R.

After the efferent arteriole, the peritubular capillary network surrounds the tubules so the kidney can reabsorb, or save, and secrete, or get rid of more, sodium, potassium, water, and so on. That's fine-tuning.

Then blood goes to the renal vein and inferior vena cava. The vasa recta are long looping capillaries in the medulla that help concentrate urine.

Two common vascular problems. One, the renal artery is occluded by a clot or stenosed by atherosclerosis, so blood flow drops, the kidney can't make urine, and worst case, renal failure. Two, basement membrane damage, as in nephrotic syndrome or some glomerulonephritis, lets things leak into urine: red cells, causing anemia; antibodies, causing poor immunity; clotting factors, causing bleeding; albumin, causing edema, since plasma loses its drawing power. Protein in urine usually means kidney damage.

Of 125 milliliters a minute of filtrate, only about 1 becomes urine. Expected output is about 60 milliliters an hour. Worry about acute kidney injury below about 30 to 40 an hour. Measure hourly output with the urimeter, the measuring box on the front of a Foley bag.

Tubular component

Proximal convoluted tubule: reabsorbs almost everything nutritionally important, glucose, amino acids, water-soluble vitamins, plus 65 to 80 percent of electrolytes, and water follows. Glucose has a reabsorption limit; in uncontrolled diabetes it's exceeded, so glucose appears in urine.

Loop of Henle and the first part of the distal tubule: lots of solute and water movement. The descending limb is permeable to water and solutes; the ascending only to solutes. By the end of the first part of the distal tubule, filtrate is as dilute as it gets.

The second part of the distal convoluted tubule and start of the collecting tubule: aldosterone acts here, retaining sodium and losing potassium; this is the last place sodium and potassium in urine change. Acid-base is also regulated here, secreting hydrogen and saving or excreting bicarbonate.

End of the collecting tubule: A D H, from the posterior pituitary, acts here to save water, so less urine. It has the final say on water.

Why location matters: the earlier a diuretic acts, the more powerful. Loop diuretics like Lasix, or furosemide, act in the loop and are most powerful. Thiazides act in the distal tubule, less powerful, less dehydration risk. Potassium-sparing diuretics act in the collecting duct.

Tubular problems: ischemia causes acute tubular necrosis, A T N, changing permeability and causing fluid, electrolyte, and acid-base imbalances. Abnormal aldosterone or A D H levels with a normal nephron also cause fluid and electrolyte imbalances.

Regulation of renal blood flow

Three mechanisms, all to keep G F R steady. Neural control, mainly sympathetic nerves to the afferent and efferent arterioles. Humoral control, hormones that constrict or dilate the arterioles; she won't ask which do which. Autoregulation, the kidney regulating its own flow, mainly through the juxtaglomerular complex, or apparatus: cells near the afferent arteriole and distal tubule monitor arteriole pressure and filtrate composition and adjust renin release. Problems here alter blood flow, urine, fluid, electrolyte, acid-base balance, and blood pressure.

Urinary tract infection

U T I is the general term. Cystitis is a bladder infection. Pyelonephritis is a kidney infection. The tract is sterile, most sterile at the kidney end, least at the urethra. Severe or chronic infection causes scar tissue, obstruction, and irreversible kidney damage, so catch it early.

Predisposing factors. One, altered urinary tract defenses. The washout phenomenon: each void flushes out organisms climbing the urethra. Void, or voiding, means emptying the bladder; she has used that word on a test. Also the bladder lining and mucus with secretory I g A. Two, immunosuppression. Three, urinary retention or obstruction. Urinary stasis, stagnant urine, is a growth medium, and you lose washout. An overdistended bladder compresses its own vessels, making the wall ischemic and vulnerable. Immobile patients on bedpans retain urine; U T I is a complication of immobility to prevent. Four, urinary reflux, urine moving backward from less sterile to more sterile areas: a cough or sneeze pushing urethral urine back into the bladder; pregnancy pressure forcing urine up the ureters; or a catheter bag held above the bladder, like hooked on a walker. Keep the bag below the bladder. Five, instrumentation: catheters, a ladder for organisms, cystoscopy, any invasive procedure. Preventing catheter-associated U T Is, CAUTIs, is a big nursing job. Six, sex: women more than men, because of a shorter urethra close to vaginal and fecal flora. Seven, fecal incontinence, contaminating the urethral area. Eight, age: weaker immunity, immobility and stasis, B P H in older men pinching the urethra, and diabetes, since glucose in urine feeds bacteria.

Cystitis signs: dysuria, painful burning urination, the best-known symptom; frequency and urgency from irritated nerve endings, like a bedpan every 20 minutes for 15 to 20 milliliters; lower abdominal or low back pain from bladder spasms; cloudy urine, from white cells and bacteria, pus; foul smell. Often little or no fever or white count rise, and kidney function is fine; people often still go to work.

Pyelonephritis signs: much sicker. Definite fever, 100 to 102, chills, malaise, nausea and vomiting; cloudy, very foul, rotten-egg urine; flank pain, mid-back wrapping to the side, more severe; elevated white count. Often starts as cystitis that climbs the ureters, so both sets of symptoms, but it can also arrive through the blood. Uncomplicated pyelonephritis doesn't impair kidney function; the patient is just miserable.

Interventions: antibiotics. Cystitis gets a short oral course. Pyelonephritis gets more aggressive treatment, I V then oral for a couple of weeks, to prevent chronic infection and permanent damage. Push fluids to use washout and prevent retention. Find and correct risk factors to prevent recurrence.

Urinary obstruction

Here, obstruction means blocked outflow of urine after it's made. Effects depend on how complete and how long. Partial or short: pain, stasis, infection. Complete, like a stone stuck in the ureter: urine backs up, distending the ureter, hydroureter, then the kidney, hydronephrosis. Pressure compresses renal vessels, causing ischemia, permanent damage, and renal failure. Causes: kidney stones, most common; B P H; and scar tissue in a ureter from chronic inflammation or recurrent U T Is.

Kidney stones

Calc means calcium; lith means stone. Names: renal calculi, urinary calculi, nephrolithiasis, urolithiasis. Any of these may be on the test.

Pathophysiology, five factors. Supersaturation: high concentration of a stone substance, calcium most often, so it precipitates, from disease like bone or parathyroid disease, or simply dehydration. Deficiency of inhibitors: kidney cells make substances that prevent stones; a genetic or acquired deficiency raises

risk. Triggers: things normally not in urine, like blood cells, pus, dead tissue, or a catheter, act as a nucleus. Example: a quadriplegic patient on bed rest lost calcium from unweighted bones, and after nine months his catheter came out encrusted with calcium. Urinary stasis: stagnant supersaturated urine precipitates more easily. And p H: some substances precipitate in acid urine, others in alkaline.

Types: calcium oxalate or calcium phosphate, by far most common, oxalate more than phosphate, related to bone disease, parathyroid disease, immobility, or dehydration. Then struvite, which is magnesium, ammonium, and phosphate. Then uric acid. Cystine is least common. Stone type guides prevention by diet, meds, or changing urine p H, so patients strain all urine to send a passed stone to the lab.

Signs. The main one is excruciating pain from two sources. One, the ureter distends behind the stone; severity relates to how quickly it distends, not how much. Two, peristaltic waves trying to push the stone cause crampy, spasmodic pain, renal colic. Location: severe flank pain radiating to the upper outer abdomen and down toward the groin, following the ureter. Pain triggers a sympathetic response: pale, cool, clammy skin, nausea and vomiting. Stones sitting in the calyces cause a dull, deep flank ache, not colic. Hematuria, because rough, spiky stones scrape the lining; even tiny stones can be very painful. Kidney function stays normal unless hydronephrosis causes ischemia, so relieve the obstruction before that.

Interventions. Pain relief is a priority: opioids like morphine, or ketorolac, Toradol, an injectable N S A I D given I M or I V. Not addictive, but short-term only because of clot risk, heart attack and stroke, and G I bleed risk. Removal depends on size. Small: push fluids, ambulate, strain urine. Too big: extracorporeal shockwave lithotripsy, E S W L. Shock waves generated outside the body are focused on the stone to shatter it. Non-invasive, may bruise, and the patient still passes fragments with pain and some bleeding. Ureteroscope: a camera with a basket to snag the stone or a laser to sludge it. A double-J stent, curled at the kidney and bladder ends, may stay a few weeks so swelling doesn't block drainage. Percutaneous nephroscopic removal, through the skin, for many or large stones; very large ones need open surgery. A staghorn calculus fills the renal pelvis into the calyces like antlers. Afterward: analyze the stone, change diet or p H, correct predisposing factors, and stay hydrated.

Renal failure: A K I versus C K D

Renal failure: kidneys can't adequately remove wastes or regulate fluid, electrolytes, and acid-base, so it affects every system. Acute kidney injury, A K I, also called acute renal failure: sudden, severe from the start, usually reversible. Chronic kidney disease, C K D: slow, gradual, milder early, lifelong and irreversible.

Pre, intra, and post renal describe the location of the cause, not time. That's a trap.

Prerenal failure: a problem with blood flow to the kidney, like a renal artery clot, massive hemorrhage, or severe heart failure with low cardiac output. No blood to filter, no urine.

Intrarenal, or intrinsic, failure: damage to the kidney itself, from autoimmune disease, ischemia, or nephrotoxic agents, meaning damaging to the kidney, like workplace chemicals, certain antibiotics, and contrast dye. Acute tubular necrosis is an example. Dividing line: poor flow with no cell damage yet is prerenal; poor flow that has damaged kidney cells is intrarenal. Prerenal is easier to recover from.

Postrenal failure: urine is made but can't get out, from obstruction, like a stone in the ureter or B P H. Back pressure compresses vessels, causing ischemia and failure. On care maps, always state why the kidney failed.

Untreated A K I can progress to C K D. C K D mostly comes from chronic systemic disease: diabetes, lupus, long-term uncontrolled hypertension, chronic kidney infections. The study guide has older and newer staging terms; she won't ask you to memorize stages or numbers. Older terms: diminished renal reserve, no symptoms; renal insufficiency, with isosthenuria and azotemia; renal failure, with uremia; and end-stage renal disease, E S R D, which needs dialysis or transplant to survive. Know the progression: G F R falls gradually from no symptoms to end stage.

Signs of renal failure

Early signs. Isosthenuria: inability to concentrate urine, so large amounts of very dilute urine, basically plasma without cells. It's the first sign of losing function, not low output. Azotemia: increased nitrogenous wastes in the blood, with no symptoms yet.

As C K D progresses: uremia, high waste levels with symptoms, plus fluid, electrolyte, and acid-base imbalances. Trap: azotemia is lab values only; uremia means symptoms.

Findings are mostly the same in A K I and C K D, because the kidney can't do its functions. Go down the list.

Wastes. Blood tests: B U N, blood urea nitrogen, from protein breakdown, affected by many things, including a high-protein diet, starvation, where the body breaks down muscle, and G I bleeds, from digesting blood protein. Creatinine, from creatine phosphate in muscle and brain, is affected mainly by kidney function and is more definitive. It changes otherwise only with a slow change in muscle mass. Lots of urea in urine is normal; urea in blood is the worry.

Uremic encephalopathy: encephalo means brain, opathy means disease. Lethargy, impaired memory and attention, confusion, then coma. Dialysis patients may be alert because dialysis removes wastes.

Platelets and clotting factors work poorly in waste, so bleeding: easy bruising, G I bleeds, so check stool for blood, and epistaxis, nosebleeds. White cells work poorly, so infection risk rises.

Peripheral neuropathy: tingling, pins and needles, restless legs, an itchy, creeping feeling worse at rest, and burning soles.

Pruritus, itching, from uremic frost: urea and phosphate crystals left on the skin when sweat dries, a white powder, as the body tries another route to excrete waste.

Also anorexia, nausea and vomiting, fatigue, and weakness. And drug toxicity, since kidney-excreted drugs build up, so lower doses or switch drugs.

Fluid. Dehydration is possible with isosthenuria in early C K D and in the recovery phase of A K I, when urine returns but can't be concentrated. More common is oliguria, scanty urine, causing fluid overload: high blood pressure, weight gain, dependent edema, and pulmonary edema with crackles.

Electrolytes. Potassium goes up: no vitamin D is needed to absorb it, and failed kidneys can't excrete it. Priority is the heart: dysrhythmias, low output, heart failure, even shock. Phosphorus goes up the same way. Calcium goes down for two reasons: its inverse relationship with high phosphorus, and no activated vitamin D, so dietary calcium isn't absorbed. Trap: despite low calcium, you don't see tetany or paresthesias. Low calcium makes the parathyroid release more P T H, which pulls calcium from bone. The bones become weak and osteoporotic: renal osteodystrophy, also called C K D-mineral bone disorder.

Result: pathological fractures with very little force, like a clavicle broken while boosting a patient up in bed.

Acid-base: metabolic acidosis. Cells constantly make acid. The lungs excrete only carbonic acid; all other acids need the kidneys. Cells don't constantly make base, so acid builds up, not alkalosis.

Anemia: no erythropoietin, so the marrow isn't told to make red cells.

Test point: three findings much more likely in C K D than A K I: anemia, because red cells live about three months and A K I usually resolves first; uremic frost, because the body must be desperate; and renal osteodystrophy, because P T H takes time to weaken bone.

Interventions and renal diet

A K I: find and remove the cause, like stopping a nephrotoxic antibiotic, and treat symptoms while the kidney recovers. C K D: damage may be irreversible, so treat the cause to slow progression, like controlling hypertension, treat symptoms, and at end stage, dialysis or transplant.

C K D symptom treatment: control hypertension, synthetic erythropoietin for anemia, calcium and vitamin D supplements for osteodystrophy, and a renal diet.

Renal diet: low sodium, so you don't retain more fluid. Low potassium and low phosphorus, because blood levels are already high. Low protein, because protein makes urea and worsens uremia. High carbohydrate, so the body doesn't break down its own protein; high fat is avoided because it's not heart-healthy. And fluid restriction, for overload. It's a hard diet to follow.

Rapid recap

A WET BED. G F R is about 125 a minute; urine about 60 an hour; worry under 30 to 40. Basement membrane is the filter. Afferent in, efferent out. Proximal saves nutrients; aldosterone acts late distal; A D H at the collecting tubule; loop diuretics strongest. Cystitis: dysuria, frequency, urgency, little fever. Pyelonephritis: fever, chills, flank pain, foul urine. Stones: mostly calcium; colic; Toradol; strain urine; E S W L. Pre, intra, post is location: flow, damage, outflow. Isosthenuria first, azotemia is labs, uremia is symptoms. Creatinine beats B U N. Potassium and phosphorus up, calcium down without tetany, osteodystrophy, metabolic acidosis, anemia. Anemia, frost, and osteodystrophy point to C K D. Renal diet: low sodium, potassium, phosphorus, and protein, high carb, restrict fluid.